

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Designed agentic AI workflows (OpenAI Agents SDK, MCP tool-calling) automating troubleshooting and reducing toil on large-scale distributed systems, resolving performance bottlenecks and reliability issues in production.
- Led design and delivery of a fault-tolerant, highly-available distributed system supporting millions of daily transactions at 99%+ uptime and sub-200ms latency, owning it end-to-end from architecture through production.
- Architected AWS serverless infrastructure (Lambda, S3, DynamoDB, API Gateway) exposing production APIs at sub-200ms latency, built observability and autoscaling with Prometheus and Grafana to keep the system stable through node loss.

QureAI

Mar 2026 – Present

ML Infrastructure Engineer

- Led migration of a synchronous service to an async, autoscaled architecture, cutting p99 latency 420ms to 140ms across 50K+ daily requests through distributed, high-concurrency design.
- Built a production observability and event-logging framework tracking latency, drift, and regressions across service versions to keep large-scale systems reliable.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and Java microservices in a distributed, fault-tolerant system for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS) with relational and NoSQL data stores.
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning across large-scale distributed data stores.
- Mentored engineers and led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams, reducing production escalations 60% in one quarter through architecture reviews.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production performance and eval tooling with rubric-based scoring, drift detection, and per-call latency benchmarking across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Go, Python, Java, C++, SQL

Distributed Systems: Fault Tolerance, High Availability, Consensus Protocols (Raft), Distributed Scheduling, Autoscaling, Prometheus, Grafana

Databases: PostgreSQL, DynamoDB, MongoDB, Relational & NoSQL Design

Cloud & Infra: AWS (EC2, Lambda, S3, DynamoDB, API Gateway), GCP (Cloud Run), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer